# The Road to the Cup Starts Here: A Basis for Predicting Playoff Success in the WHL

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***Abstract*—In this paper, we examine the impact of regular season performance on Western Hockey League (WHL) playoff success. We analyze head-to-head records, player statistics, and team statistics to construct probability matrices detailing the likelihood that team $i$ beats team $j$ in a playoff game. Using stochastic simulations for all playoff series between 2013 and 2024, we evaluate team performance. Finally, we enhance our model by incorporating machine learning techniques to optimize the probability matrices, refining predictions and improving playoff success forecasts based on regular season data.**

## I. Introduction

FOUNDED in 1966, the Western Hockey League (WHL) is one of the best developmental hockey circuits in the world. Alongside its Canadian Hockey League (CHL) counterparts, the Ontario Hockey League (OHL) and Quebec Maritimes Junior Hockey League (QMJHL), the WHL holds a major junior designation, placing it at the peak of Canadian junior hockey. With 22 teams spread across 4 Canadian provinces and 2 U.S. states, the WHL boasts over 127 players named to National Hockey League (NHL) opening night rosters for the 2024-2025 season. The league is a junior hockey powerhouse and the pinnacle of the sport for skaters older than 16 and younger than 20.[1]

### A. WHL Regular Season and Playoff Structure

In 2018, the WHL reduced the regular season from 72 games to 68 games. Each of the two WHL conferences, Eastern and Western, consists of 11 teams. Teams in the Eastern Conference play one regular season game against every team in the Western Conference. The Western Conference is divided into the U.S. Division (6 teams) and the B.C. Division (5 teams), while the Eastern Conference is split into the Central Division (5 teams) and the Eastern Division (6 teams). The number of games a team plays against divisional opponents varies depending on the size of their division, but the total number of in-conference games for each team is 57. Therefore, the 68 game season is broken into: 11 inter-conference games and 57 in-conference games.

Every year, the WHL playoffs begin in mid-to-late March following the conclusion of the regular season, featuring the top eight teams from each conference. The top three teams in each division (six total teams, as each conference two divisions) earn automatic playoff spots, while the remaining two wild-card berths are awarded to the next highest-ranked teams in the conference, regardless of division. The playoffs follow a best-of-seven series format, with re-seeding after each round to ensure the highest-seeded team always faces the lowest-seeded remaining opponent. This reseeding posed a new challenge away from the traditional tournament simulation structure. Further, this re-seeding format emphasizes regular season performance, rewarding top-seeded teams with more favorable matchups as they advance. Teams progress through the Conference Quarterfinals, Semifinals, and Finals before competing for the WHL Championship. The winner earns a spot in the Memorial Cup, a prestigious tournament featuring champions from across the entire CHL (a battle of the best teams across the best leagues in Canada, so to speak).

### B. Project Intent and Motivation

The WHL's age cap and free agency structure make it an intriguing case for studying playoff predictions. Teams vying for Memorial Cup qualification, achieved by winning the WHL, often trade future draft picks at the deadline. These trades bring in older, more experienced players to strengthen their rosters and boost their chances of a championship run. This is very similar to the behavior of championship contenders across the majority of different Sports leagues - sacrifice future draft capital to try to capture your "Championship Window". However, the WHL's constant roster turnover, driven by player graduations and trades, along with the limited number of inter-conference games, makes outcomes highly unpredictable. As a result of these factors, the roster composition of each team on a season-over-season basis is highly variable, making recent season success an unreliable predictor of future returns. Therefore, repeat champions are rare, and traditional models implemented in other sports leagues with less roster turnover often fall short. This project leverages historical data and advanced modeling techniques to develop a more reliable approach to predicting playoff success in the WHL.

## II. Model Development and Description

### A. Motivation

With our reasoning for trying to model the WHL established, we wanted to examine different methods that could effectively model and hopefully predict playoff success. Several different methods for tournament simulations exist - from Markov Chains to a variation of Mosteller's Baseball Model. However, we turned to our classmates' work in the *AM215 GitHub Repository* as an optimal solution to which we try and approach.

[1] A handful of players qualify for exceptional status, meaning that they can play in the WHL at the age of 15. Each team can have 3 players that are 20 years old. Upon turning 21, that player can no longer compete for their team.

The code from the *GitHub Repository* lent encouragement towards a variation of the Elo model. We will discuss this more in depth in later sections, but we did not just approach our game simulations as simple Elo based probabilistic matchups. Instead, we drew inspiration from the *TennisEloEnhancement* library used in the Repository, which adds additional dynamic nuance specific to Tennis with analogous characteristics to our hockey adjustments.

Before diving deeper into the heart of our basis model, the adjustment to the typical Elo model, we must first lay the groundwork for our typical tournament simulation method: The Monte Carlo Tournament.

### B. The Monte Carlo Tournament

At a high level, the Monte Carlo Tournament projection simulates the outcomes of a tournament based on a series of probabilistic inputs. Determining the correct probabilistic inputs that drive the result of the Monte Carlo Simulation, however, is the key part to any successful simulation. Each simulation involves determining the winner of each game within a series, iterating on these stochastic drivers until a team wins the series. The winner of the series then advances to the next round, where the process repeats with a different competitor. Eventually, a winner is crowned. This simulation is then repeated N number of times, creating a distribution of outcomes that is indeed robust and is intended to account for variability and uncertainty, assuming the probabilistic inputs are well selected.

Let's examine the Monte Carlo for the case of the WHL. For $n = 22$ teams in the WHL, the pairwise probability matrix, which details the probability, $p_{ij}$, that team $i$ defeats team $j$, can be denoted as follows:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}.$$

Let $T_i$ and $T_j$ represent two competing teams in a playoff series. Let $p_{ij}$ be the probability of team $T_i$ winning a single game against $T_j$. For each game in the series, sample $u \sim \mathcal{U}(0, 1)$.

First, the outcome of each game can be defined as follows:

$$W_{\text{game}} = \begin{cases} 1 & \text{if } u \leq p_{ij} \\ 0 & \text{if } u > p_{ij} \end{cases}$$

Then, we accumulate wins for each team over the best-of-seven series:

$$W_i = \sum_{k=1}^{N} W_{\text{game},k}, \quad W_j = N - W_i,$$

where $N$ is the total number of games played until one team wins 4 games.

If at any point during the series, a team wins 4 games (i.e., $W_i = 4$ or $W_j = 4$), the series ends early. Thus, $N$ is the number of games played until the first team reaches 4 wins:

$$N = \min(7, \text{first round where } W_i = 4 \text{ or } W_j = 4).$$

Finally, the series winner can be determined:

$$W_{\text{series}} = \begin{cases} T_i & \text{if } W_i \geq 4 \\ T_j & \text{if } W_j \geq 4 \end{cases}$$

### C. Determining the Probability Matrix

The most challenging aspect of simulating the WHL playoffs in Monte Carlo fashion is determining how to properly fill the pairwise probability matrix. In the following subsections, two separate formulations for creating this matrix will be explained:

*1) A Head-to-Head Formulation:* First, we will initialize a naive model that predicts each team's probability of winning the WHL by strictly using their regular season head-to-head matchups. Given the fact that the WHL has 22 teams that each play 68 games, we believed this to be an acceptable first approach towards determining the probability matrix. We scraped and pieced together a data frame of the results of each of every single regular season game from the WHL from 2013-2024. Please note: the 2020 and 2021 seasons were canceled entirely due to the COVID pandemic, and the 2022 season was severely affected due to game restrictions and cancellations. Because of this, our analysis will exclude those years.

After putting the data frame together, we iterated through every season, and within each season put together each team's W/L probability based on every single head-to-head result. However, we know that hockey has three different loss outcomes that should not be weighted equally: a regular time loss, an overtime loss, and a shootout loss. Conceptually: if a team loses to another in regular time, this should carry more weight than if a team lost in overtime. This is because an overtime loss dictates that the teams are likely remarkably even in ability and performance, as they went 60 minutes and finished even. On the other hand, a regular time loss indicates that 60 minutes was enough to decide a winner. The most extreme version - a shootout loss - indicates that 60 minutes of regular time AND an overtime period were not enough to declare a winner. As such - the loser of an overtime game (and even more so in a shootout loss), would not be as punished as a team that lost in regular time. Evidently, the corresponding winning team in an overtime or shootout game would not gain as much as winning in regular time.

To account for this difference, we define a parameter $total\_losses$ that acts as a part of the denominator for both teams and the numerator in the changes to the losing team's matrices.

$$total\_losses = losses + ot\_losses + so\_losses$$

$$win\_total\_loss\_matrix = \frac{wins}{wins + total\_losses}$$

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Examining the win/loss probability matrix, we see that there are far too many definitive probabilities (at 1 or 0). Though this does not constitute a category error, we realize that this is not reality as a team always has a non-zero chance to win a game (especially in a competitive league such as the WHL). Therefore, we decided to add two different factors to our "naive" head-to-head model - transforming it into a "more intelligent" head-to-head model.

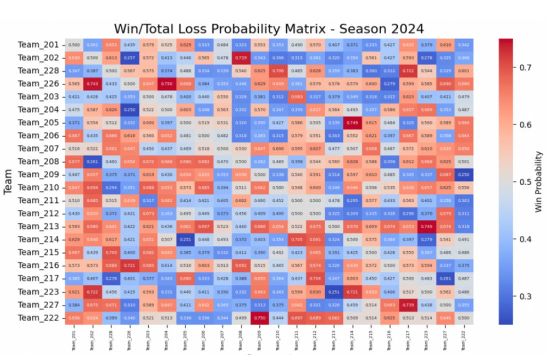
Firstly, we add a noise constant to every single element of the probability matrix. We model noise in this initial model with a stochastic factor that is contingent upon the number of head-to-head games played between rival teams and each team's total number of wins and losses. We establish an inverse relationship here - there more games played between two teams, the less significant the random factor that impacts their matrix probability will be. Therefore, the opposite is true - meaning if two inter-conference teams only played twice, the delta factor will be more significant. This is represented in the formula:

$$randomness\_scale = 0.5 * \frac{1}{\sqrt{games\_played\_between + 1}}$$

This randomness scale is then multiplied by the random delta factor. Significant thought went into this randomness implementation, as we believe hockey to be the sport perhaps most influenced by random events to determine outcomes of games. For example, one weird bounce of the puck off of the wall could create a sure-thing scoring chance for a team, to no fault of the unlucky team.

Secondly, we wanted to address the extremity of the probability matrix. The naive head-to-head algorithm was simply putting far too much weight on matchups that only occurred twice in a season. Though the randomness factor certainly handled a significant portion of this, we felt it necessary to scale the calculated probabilities across a fixed range. We selected the range of $[.25, .75]$ as no team should track to less than a $\frac{1}{4}$ chance of winning of a given night, given the relative parity of talent across the league. To do this, we define a helper function that reweighs a range of numbers into a specific range with a normalization process.

Following both the randomization and scaling processes, our resulting probability matrix for 2024 yields:

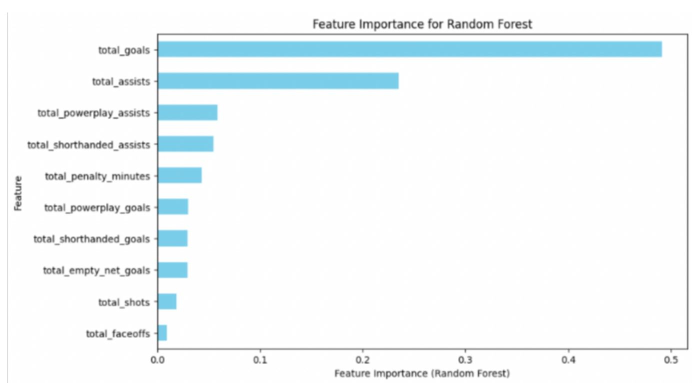



Now with each team's $i$'s probability of winning a game against team $j$, we will simulate each game and each series, according to WHL playoff rules, in order to identify the strongest teams. We will discuss these results in the analysis section.

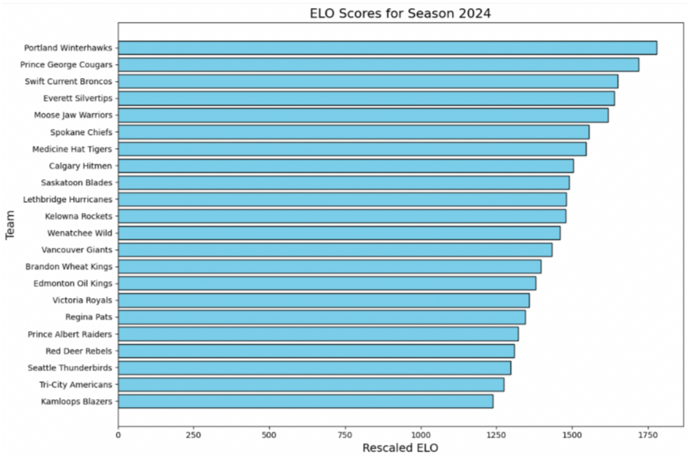
*2) A Statistics-Driven Approach:* Next, we will take a statistics-driven approach, using machine learning techniques to identify which trends are associated with winning teams. Standings in hockey operate slightly differently than just a Win/Loss record - teams are ranked by total points earned. A team gets three points for a win, one point for an overtime loss, and zero points for a regular time loss. Therefore, making regular season team points earned per game our target variable, we will analyze the importance of each statistical predictor through a Random Forest Model.

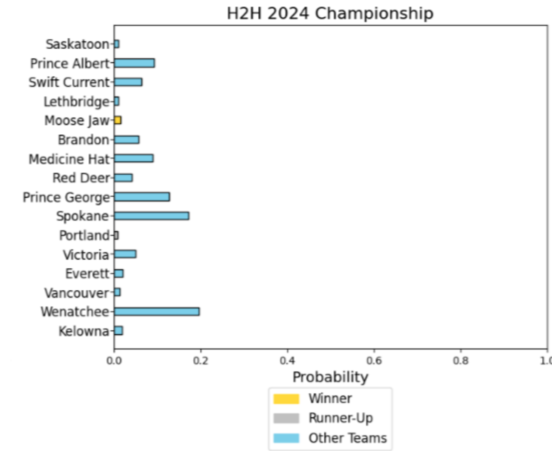
We were able to scrape and create a data frame of key team metrics for every team, across all seasons discussed earlier. We viewed the natural split in our data (2013-2019), then (2023, 2024) as a great way to decide what to use as our training and test data. As such, we trained our Random Forest model on the data from (2013-2019), and used these coefficients to attempt to predict and build to appropriate Elo scores for our test period (2023, 2024).

Our coefficients would then apply to the metrics that include totals for: points, goals, assists, power play stats, shots, and face offs. A point in hockey is either a goal or an assist. For example, if a player has 10 goals and 5 assists on the season, then they finish with 15 points. Therefore, it seems redundant to have both point totals as well as goal and assists totals. We do want to examine the effects of goals and assists separately, however. Therefore, we will remove the effect of the point statistics while leaving the other two. Using these predictors, we will assign each team a performance score that reflects their overall strength based on regular season metrics. This performance score will be our build of creating each team's Elo ratings!




This process is not entirely straightforward: we first use the normalized importance given to us from the Random Forest for each feature. We then normalize each team's totals for every statistic, and multiply these by that corresponding statistics normalized importance weight. We then have every team's, in every year, raw score, which we use a *MinMaxScaler* to convert to a standard Elo scale (between $1100 - 1800$).




Our Elo prediction method here is statistically driven, and heavily inspired again by the *AM215 GitHub Repository*. Specifically, its use of scaling different court elements such as: clay, grass, carpet, and hard with different weight factors depending on their importance.

Our calculated Elo scores will then be transformed into probabilities through a logistic function, creating a comparative framework for head-to-head matchups. We will use this framework for running our tournament simulation once more. The standard Elo logistic regression to convert from Elo rating to head-to-head probabilities is given by:

$$P(A\_beats\_B) = \frac{1}{1 + 10^{(Elo\_B - Elo\_B)/400}}$$

$$P(B\_beats\_A) = 1 - P(A\_beats\_B)$$

We apply the same scaling factor to add realism to our model, keeping any one game win outcome probability between $[.25, .75]$. These equations yield a Elo driven probability win/loss heat map for 2024 (similar in structure to those shown prior).

## III. Analysis

### A. Model Analysis

To begin our analysis of our two models, let's first return back to our "more intelligent" head-to-head formulation. We last left off at the win/total loss probability matrix that includes both a dynamic randomness factor as well as a scaling factor to our realistic range of probabilities.

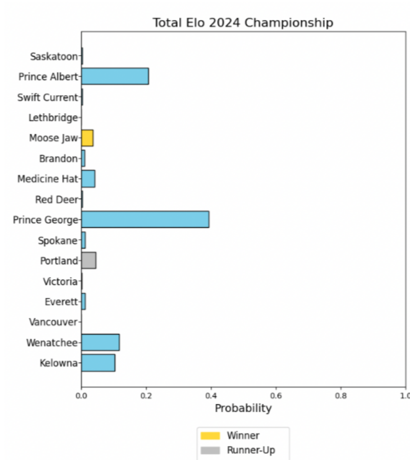
Within each tournament, every game is decided by stochastic process, which determines the winner of the rounds, which determines the winner of said tournament. In order to run our analysis, we run $10,000$ simulations of different tournaments. This results in the championship prediction for 2024 of:

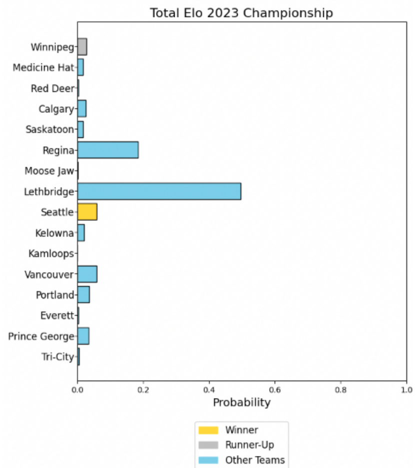
As evident, the "more intelligent" head-to-head matchup has an extremely difficult time predicting the actual outcome of the tournament, at least in 2024. Teams that performed extraordinarily well in the regular season also performed well in our prediction - such as Wenatchee and Spokane. However, this result should not be surprising - aside from our dynamic randomness factor and scaling efforts, only head-to-head results matters in determining this model's predictions.

To validate the model against of the years, we see that the model only predicts the correct winner *once* in the nine years of valid predictions. This came in 2014, where Edmonton was such a strong regular season favorite that their probability of advancing to the semifinals was greater than $0.8$.

As we expected, even a "more intelligent" head-to-head model was going to struggle to predict playoff outcomes based on regular season success alone. In all sports this would likely be the case, but especially in a sport as "up for grabs" as the hockey playoffs are renowned to be. Let's turn our attention back to the statistical Elo approach.

Similar to the head-to-head model, we run $10,000$ simulations of tournaments based on our Elo generated head-to-head win/loss probability matrix. Different than the head-to-head, however, was the machine learning approach inherent to our model. Therefore, let's only examine the test data (2023 and 2024), as defined in the model description.






Examining the results for 2024, we first see that again, the model is not a very strong predictor. This is an initially frustrating result - but as we will continue in the discussion section, the WHL is an especially difficult league to predict. Teams can lose their best players to the NHL at any moment. Further, the impact of the severe trade deadline can change the complexion of a team in the middle of the regular season (for better or for worse) so their rating may not reflect how strong they are into the playoffs.

However, there is a noticed improvement over our "more intelligent" head-to-head model. Though still not very high, the predicted winner Moose Jaw is 2x more likely to win in this model than before. An even greater improvement, we see that the runners-up, Portland, improve their odds from near zero to one of the best in the league.

Examining across the rest of the years, we see improved predictions nearly across the board. In 2023, this also holds true - though we are showing the result now as to its relevance in the next section. This gives us hope in that the statistically driven Elo prediction model may hold merit in starting to crack the unpredictable WHL. We wanted to expand on it further...

### *B. Sensitizing the Model and Continuation*

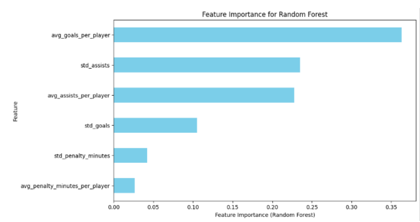
The process of converting from Elo ratings to head-to-head win probabilities is near fixed... this is widely accepted and used in practice. Therefore, we want to sensitize on what happens when we use different statistics in our Random Forest model to create the Elo rankings.

Instead of looking at team total stats, we wanted to look at the conjunction of player per game averages. This alone would not be overly different, but we also now have access to the standard deviation of these stats across games - which allows consistency to feed into a factor that can predict playoff success, an exciting prospect!

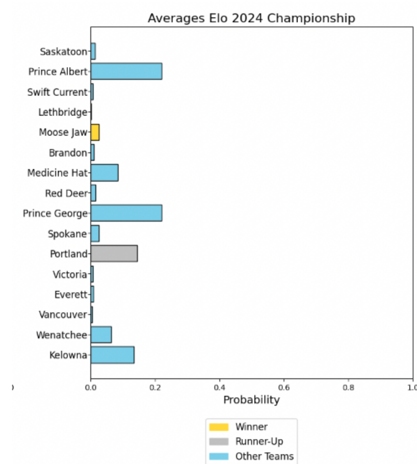
The statistics that we were able to scrape and create in a data frame were averages and standard deviations in games for each of the following stats: plus/minus, assists, goals, points, penalty minutes. For the same reason as before, we did not include points. The (+/-) metric indicates how many goals net the player is on the ice for. For example, if a player's team scores 23 goals while he is on the ice across the whole season and the other team scores 13 goals, then his (+/-) would be 10.

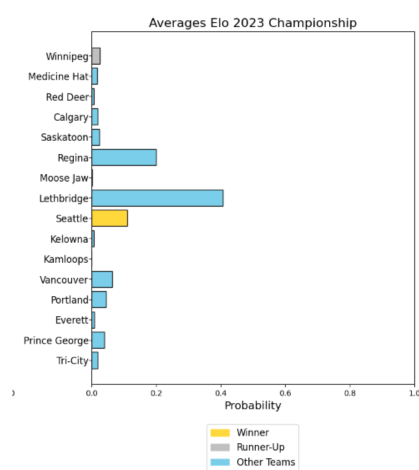
Therefore, it entirely makes sense that this metric is the leading indicator for season success is how successful the players are when they are on the ice, on average. As a result of this, we want to take more of a systematic approach of players' stats leading to team success.

Running the Random Forest regression on the same test period (2013-2019) yields the importance factors below:




The most surprising result here is that the average goals per play is more impactful than the standard deviation of goals. Perhaps the ability to score a lot of goals indeed outweighs the ability to score consistently. Plugging these coefficients and running the same process as before yields the results for 2023 and 2024 (the test set).



We see that again, the model is not very good at predicting the league winner. Across all model variations, we see that Moose Jaw was very difficult to predict. However, in the averages variation of the Elo model, we see a significant increase in the runner-up for 2024 (Portland), as their odds of victory nearly 3x. Looking at the other portion of the test set in 2023, we see an increase in prediction for both the runner up (Winnipeg) and a significant boost to the league champions in Seattle.

Though the models do not perform overly well, this was expected given the degree of randomness and change in the WHL. We are building momentum towards a more accurate prediction model (with the averages Elo model being the best thus far), which we plan on expanding much further for our final project.

## IV. Discussion

Our results are not overly encouraging, but the tuning of our Elo model leads us to believe that we are on the right track. Further, note that all of our statistics that feed into the Elo rating are offensive focused stats. We still need to look at defensive stats, including the Goalie statistics file that we have scraped and begun to put together. Our final project will indeed have an optimized linear combination of a scaled offensive and defensive score to build out the Elo rating.

There are several factors that make the WHL particularly hard to predict, but we feel confident that we can tackle them head-on into the final project:

1) **Lack of Inter-Season Continuity:** Because of the nature of the journey league, there is extremely high roster turnover. There is a strict age restriction, where players are not allowed to play at all by twenty-one, and only a certain number of players can be twenty on a roster at a time. Therefore, most of a team's best players will likely leave between season to join the NHL or other professional leagues. In order to tackle this issue in our final project, we are going to look at the influx of players that leave for the NHL and other professional leagues. We will attempt to look at the attributes that define these players (size, statistical production) and attempt to see if we can predict this between seasons for teams. Further, WHL players will now be allowed to play college hockey (as of November 7, 2024) - so we can begin to try and predict this as well.
2) **Lack of Intra-Season Continuity:** Following similar suit to the prior point, the forced age restriction creates significant roster turnover while *in*-season. Teams are often forced to trade some of their best assets due to roster management around the age cap rules to remain eligible. There is a large amount of game theory around this - for example, if a team is struggling they may want to take expiring, older contracts off contenders' hands in exchange for a young prospect who may help them down the road. These trade deadline dynamics are extremely unique, and we will inspect the impact of the trade deadline on team success in our final project. To do this, we will analyze the changes in team composition before and after the trade deadline, focusing on the acquisition of veteran players or key positional upgrades. By tracking team performance metrics such as points per game, goal differential, and win/loss records before and after the deadline, we can quantify the effect of roster changes on playoff outcomes. Using this information, we will adjust our model to account for the influence of trade activity on a team's probability of success in the postseason. This will allow us to refine our predictions further, incorporating the added complexity of strategic roster changes during the season.
3) **Model Validation:** Perhaps the trickiest aspect is going to be validating our model as we make it more specific without running the risk of overfitting. For example, the way that we currently evaluate our model (against actual results) is a good proxy, but it would be best to find betting odds, convert these to probabilities, then compare that MSE. We will look into doing this for our final project, though we believe this data may prove very difficult to come by.
4) **Dataset:** Lots of manual errors went into scraping the WHL website in order to create the datasets used in this project. Because of this, it is very possible that we are missing key stats that could impact the way that the Elo ratings are created. For example, something such as hits (a defensive stat) may indeed be predictive to a team's strength.

## V. Summary

We were originally enticed towards predicting the WHL because of its perceived difficulty and variability, and we found this to be every bit as true as first thought. We demonstrated that a head-to-head model, even with a dynamic randomness factor and realistic scaling, falls woefully short of being considered useful. Shifting towards a Machine Learning approach to building to Elo ratings, we saw that the predictions for the test data became more reasonable, but not yet practically useful. Finally, our last iteration (for now) of building to Elo ratings using in-game averages and variability yields the most useful model for predicting playoff outcomes in the league.

Now that we have established a roadmap and baseline Elo prediction model to work off, we believe that adding nuances, such as trade deadline and off-season modeling, into our final project will lead to a more comprehensive predictive model for the league. Expanding on our Elo model to include defensive metrics should provide a more comprehensive model, and sensitizing across different linear combinations of these two ratings should yield stronger results. As we meet a larger scope and ambition of the project into the final stage, we will remain diligent as to not overfit by constantly validating our training data against any historic odds or results.

## Appendix A
## Python Notebook

Click here to access our Python supplement. We have cultivated a notebook that mirrors the progression of our paper and justifies our graphics.

## Acknowledgment

Thank you to Professor Brenner, TF Elle Weeks, enhanced CA Erik Wang, and our respective sections for their feedback and guidance throughout the development of this project.

*We would like to give a necessary attribution to ChatGPT, which helped us in random spots in our notebook. We derived all of the logic and initial code ourselves, but ChatGPT lent us a hand with as we sought to create powerful visuals and debug persistent syntax errors.*

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